STANDARD COLLEGE NTUNGAMO

PRE REGISTRATION 2026

S6 PHYSICS PAPER 1

TIME : 2 HOURS

INSTRUCTIONS: Attempt all questions

SECTION A : PARTICLES

ITEM 1

At Mulago National Referral Hospital, a team of physicists is training new interns in the use of radioactive materials for diagnosis and treatment. Three radioactive sources are presented.
-source A emits radiations which are stopped by a sheet of paper.
-source B emits radiations which are stopped by the thin sheet of aluminum.
-source C emits radiations which are stopped by the lead.

TASK

a) Identify the type of the radiations in each source and state two properties of each
b) State any four safety precautions these interns should follow which handling radiations
c) The half life of Thorium-234 is 5.45 years. Determine the activity of 2g of Thorium.

Hint (Avogadro constant is 6.02×10^{23} per mole)

SECTION B : FORCE AND MOTION

ITEM 2

At a research station on Lake Victoria, a group of physics students observes scientists studying small ripples moving across the lake surface.

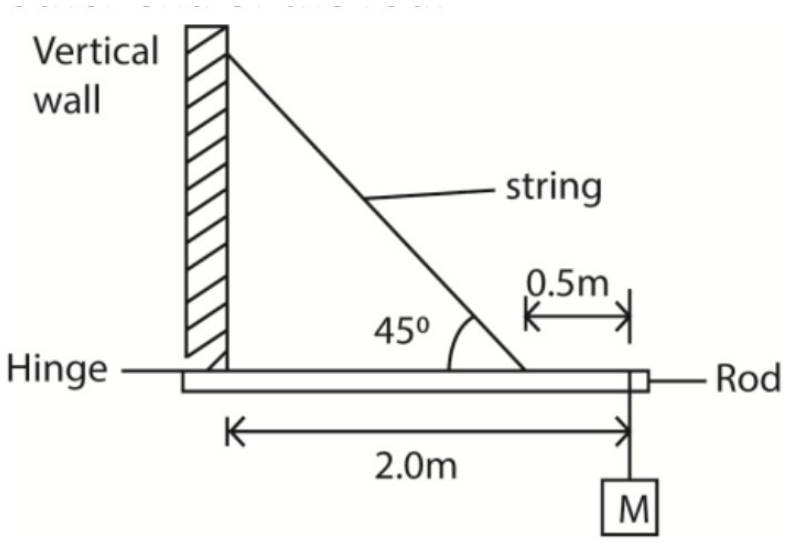
One scientist writes on a board:

$$v^2=\frac{(\lambda g)}{(2\pi)}+\frac{(2\pi\gamma)}{(\lambda\rho)}$$ where g is the acceleration due to gravity

and explains that this equation relates the velocity, v of the wave of wavelength, λ, on the surface of a liquid of surface tension, γ and density, ρ. A student notices that each quantity in the equation is written in terms of other quantities which the scientist refers to as the dimensions of the physical quantities.

Meanwhile, another scientist demonstrates a motor pulling a sensor along a horizontal rod. The motor exerts a steady push on the sensor, causing it to move at a constant speed. He explains that the rate at which the motor transfers energy to the sensor, P depends on both force applied, F and velocity of motion, v.

Nearby, engineers are installing a horizontal rod of negligible mass hinged to a vertical wall to support a 20 kg measuring device. The rod is held in place by a string attached 0.5 m from its free end as shown below.



As they adjust the system, one engineer emphasizes that for the rod to remain stable, the turning effects about the hinge must balance.

The string used in the setup can safely withstand a maximum tension of 500 N.

HINT

Acceleration due to gravity, g = 9.81 ms^{-2}

TASK

(a) Referring to the equation written by the scientist on the board, explain what is meant by the term used by the scientist , and use the given relationship to show that the equation for wave velocity is valid.

(b) From the demonstration of the motor pulling the sensor along the rod, derive the relationship between P, F and v.

(c) Based on the engineer's explanation during installation, state the principle that ensures the rod remains stable about the hinge.

(d) Using the support system described in the scenario:

(i) Determine the tension in the string.

(ii) Find the reaction at the hinge.

(iii) Calculate the maximum additional mass that can be added to the 20 kg device without exceeding the safe tension of the string.

SECTION C: ENERGY

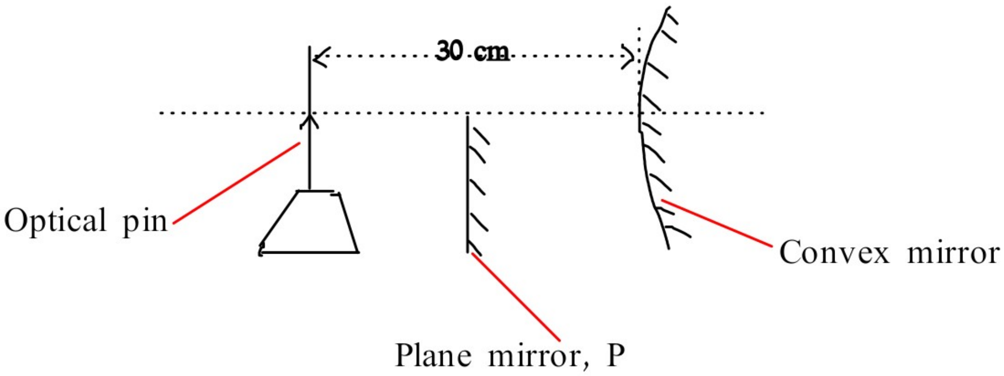
ITEM 3

In a physics laboratory in Kampala, a student sets up an optical experiment using a convex lens of focal length 15 cm to determine the focal length of a convex mirror. He also placed a plane mirror P to a position 20 cm from the optical pin such that the image of the object pin in P coincides with that formed by the convex mirror.

In a separate activity, the student studies how the particles in waves move as the wave travels through a medium. He also found out that when a plane wave traverses a medium, the displacement of particles is given by:

$y=0.01 \sin 2\pi(2t-0.01x)$,

where distance and displacement are in metres and time in seconds. The student compares how particles vibrate in the wave and measures the separation between two particles as 50 cm.



TASK

(a) Describe how the student can experimentally determine the focal length of the mirror.

(b) Calculate the focal length of the mirror.

(c) Differentiate between the two particle motions

(d)Using the wave equation provided:

(i) frequency of the wave

(ii) Determine the wave velocity.

(iii) Calculate the phase difference between the two particles.

SECTION D: CHARGES AND FIELDS

ITEM 4

In a rural home, an electrician is installing a solar backup system to store electrical energy for use at night. He uses a 60 μF capacitor connected to a 120 V battery. While testing the system, the charged capacitor is accidentally connected across the terminals of an uncharged 20 μF capacitor, causing energy redistribution.

At the same home, a metal rod is fixed vertically on the roof and connected to the ground using a thick conducting wire is installed to protect the building during thunderstorms. A physics student observing the installation becomes curious about how charge behaves in conductors and performs an experiment using a hollow metal container.

TASK

(a) Using the capacitor setup described, explain what is meant by capacitance and state its S.I unit.

(b) (i) Using the values of the capacitors and battery given in the scenario, determine the final potential difference across the capacitors after they are connected.

(ii) Using the initial conditions and final state described, calculate the initial and final energies stored in the capacitors and suggest a reason for the energy difference observed.

(c) Referring to the metal rod and wire described, explain how this setup protects the house from damage during a thunderstorm.

(d) Describe how the student can verify where charge is located on the metal container.

END